

The Fundamental Theorem of Calculus (5.3)

Zero's arrow flies in a straight line with velocity $v(t)$, during the time interval $[a, b]$. What is its position at time T , where $a \leq T \leq b$?

$$s(T) = \int_a^T v(t) dt$$

Fundamental theorem of calculus, part 1

Let f be a continuous function defined on $[a, b]$,
and let

$$g(x) = \int_a^x f(t) dt$$

for $a \leq x \leq b$. Then $g'(x) = f(x)$ for all $a < x < b$.

Example: Let

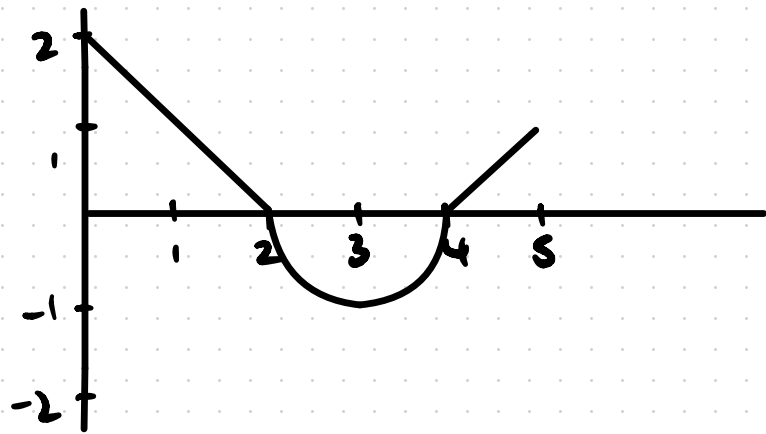
$$g(x) = \int_0^x 2t \, dt$$

What is $g'(x)$?

Example: Let

$$g(x) = \int_0^x f(t) dt$$

where f is the function



Example: Let

$$g(x) = \int_2^{x^2} t^3 dt$$

What is $g'(x)$?

Example: Let

$$g(x) = \int_x^1 e^{t^2} dt$$

What is $g'(x)$?

Example: Let

$$g(t) = \int_{\sqrt{t}}^{2t} \sin x \, dx$$

What is $g'(t)$?

Zero's arrow flies in a straight line with velocity $v(t)$, during the time interval $[a, b]$. What is its position at time T , where $a \leq T \leq b$?

If we assume $s(a) = 0$, then

$$s(T) = \int_a^T v(t) dt$$

Fundamental Theorem of Calculus, part 2.

Let f be a continuous function defined on $[a, b]$.

Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f .

Example: Evaluate

$$\int_{-1}^1 (1-x^2) dx$$

Notation: A common shorthand for

$$F(b) - F(a) \quad \text{is} \quad F(x) \Big|_a^b$$

Example: Evaluate

$$\int_0^{\pi/3} \sec^2 \theta \, d\theta$$